Pearson Edexcel International GCSE

Mathematics A

Modular Unit 1

June 2022 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

29 adaptive question sections   100 marks

Selected from Linear Higher Papers 1H and 2H

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Elite IGCSE Mathematics • eliteigcse.com

L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each question section keeps the exact source demand beside practical working space. Source pixels are retained once; long source panels continue on numbered pages.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. This book contains the demands owned by Unit 1; the selection contains 100 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Candidate source assets are retained for private teaching use only.

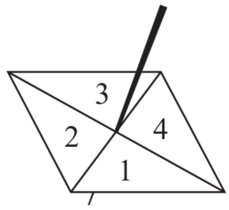
Contents

Question 01

4MA1-1H Q2 Q2

MODULAR UNIT 1
4 marks

2 Here is a biased 4-sided spinner.



The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

Number	1	2	3	4
Probability	0.26		0.18	

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Ravina is going to spin the spinner a number of times.

Ravina works out that an estimate for the number of times the spinner will land on 3 is 45

Work out an estimate for the number of times the spinner will land on 4

....

Working space for Question 01

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 02
4MA1-1H Q5 Q5(a)

MODULAR UNIT 1
2 marks

5 (a) Expand and simplify $(n-6)(n+4)$

....
(2)

WORKING SPACE

Question 03

4MA1-1H Q5 Q5(b)

MODULAR UNIT 1
3 marks

(b) Solve $2x - 3 = \dfrac{3x - 5}{4}$

Show clear algebraic working.

$x =$ ....
(3)

WORKING SPACE

Question 04

4MA1-1H Q9 Q9(a)

MODULAR UNIT 1
1 marks

9 (a) Simplify $8 \times (4t)^0$

....

(1)

WORKING SPACE

Question 05

4MA1-1H Q9 Q9(b)

MODULAR UNIT 1

1 marks

$x^6 \div x^{-5} = x^p$

(b) Find the value of p

$p =$ ....

(1)

WORKING SPACE

Question 06

4MA1-1H Q9 Q9(c)

MODULAR UNIT 1

2 marks

(c) Simplify fully $(2k^2m^4)^3$

....

(2)

WORKING SPACE

Question 07

4MA1-1H Q11 Q11(a)

MODULAR UNIT 1
2 marks

11 (a) Factorise $9x^2 - 4y^2$

....

(2)

WORKING SPACE

Question 08

4MA1-1H Q11 Q11(b)

MODULAR UNIT 1
3 marks

(b) Express $\dfrac{7}{8} - \dfrac{x+3}{4x}$ as a single fraction in its simplest form.

....
(3)

WORKING SPACE

Question 09

4MA1-1H Q12 Q12(a), Q12(b)

MODULAR UNIT 1

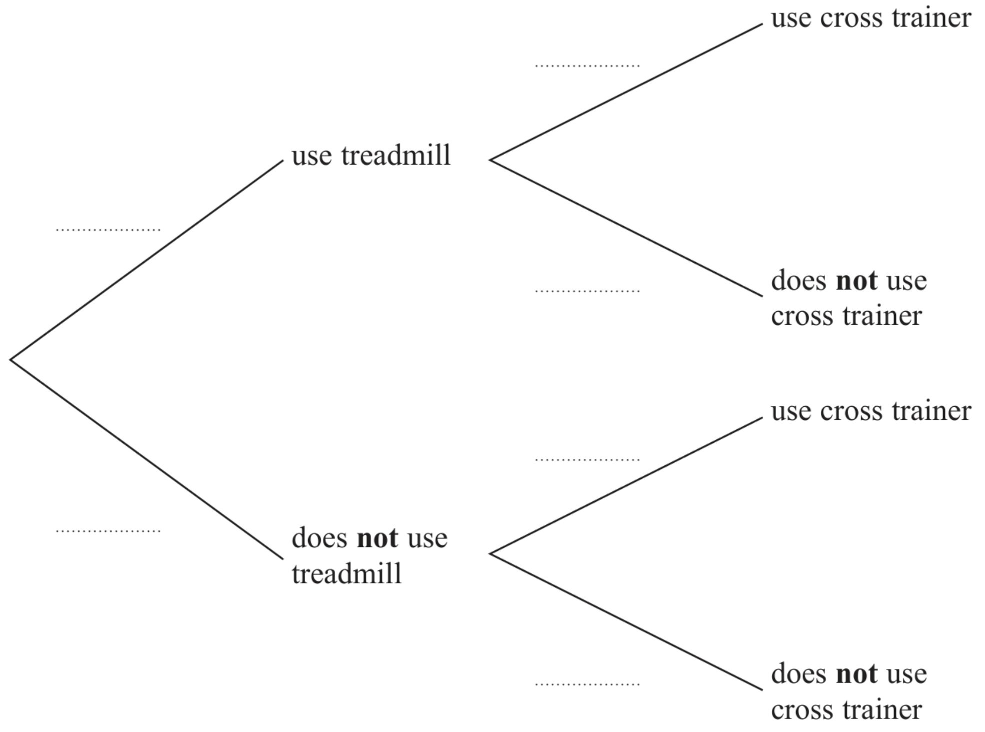
4 marks

12 Rudolf goes to the gym.

The probability that he will use the treadmill is 0.8

When he uses the treadmill, the probability that he will use the cross trainer is 0.3
When he does **not** use the treadmill, the probability that he will use the cross trainer is 0.6

(a) Complete the probability tree diagram for this information.



(2)

(b) Work out the probability that Rudolf uses both the treadmill and the cross trainer.

....

(2)

Question 10

4MA1-1H Q15 Q15

MODULAR UNIT 1
4 marks

15 Abraham is going to play a computer game.
Abraham can win the game, draw the game or lose the game.

For any game that Abraham plays

the probability that he wins the game is 0.3
the probability that he draws the game is 0.5
the probability that he loses the game is 0.2

When Abraham wins a game, he scores +10 points.
When Abraham draws a game, he scores 0 points.
When Abraham loses a game, he scores −5 points.

Abraham plays 3 games and the points he scores in each of the 3 games are added together to get his total score.

Work out the probability that when he has played 3 games his total score is 0 points.

....

WORKING SPACE

Question 11

4MA1-1H Q16 Q16

MODULAR UNIT 1
3 marks

16 Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - \left(\sqrt{2}\right)^5 = 2\sqrt{32} + 12$

Show your working clearly.

WORKING SPACE

Question 12

4MA1-1H Q18 Q18

MODULAR UNIT 1
5 marks

18 Here is triangle ABC

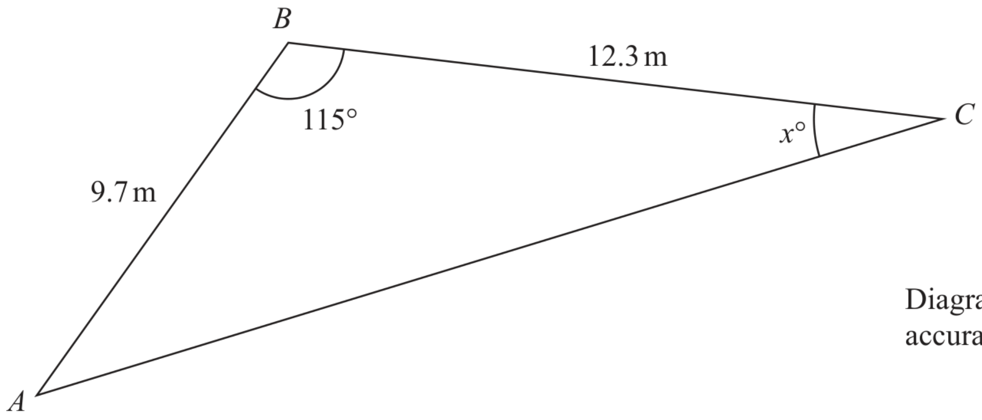


Diagram **NOT** accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$ ....

Working space for Question 12

Continue your method

UNIT 1
5 marks

WORKING SPACE

Question 13

4MA1-1H Q19 Q19

MODULAR UNIT 1
5 marks

19

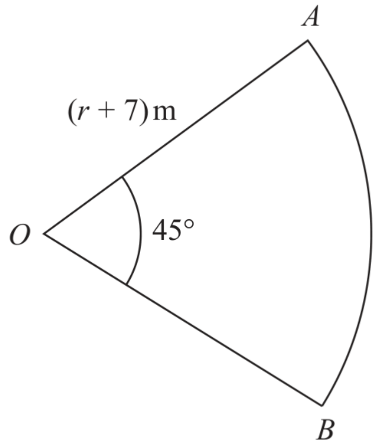


Diagram **NOT** accurately drawn

OAB is a sector **S** of a circle with centre O and radius $(r + 7)$ metres.
Angle $AOB = 45°$

A circle **C** has radius $(r - 2)$ metres.

The area of sector **S** is twice the area of circle **C**

Find the value of r
Show your working clearly.

Question 19 continued.

$r =$ ....

Working space for Question 13

Continue your method

UNIT 1
5 marks

WORKING SPACE

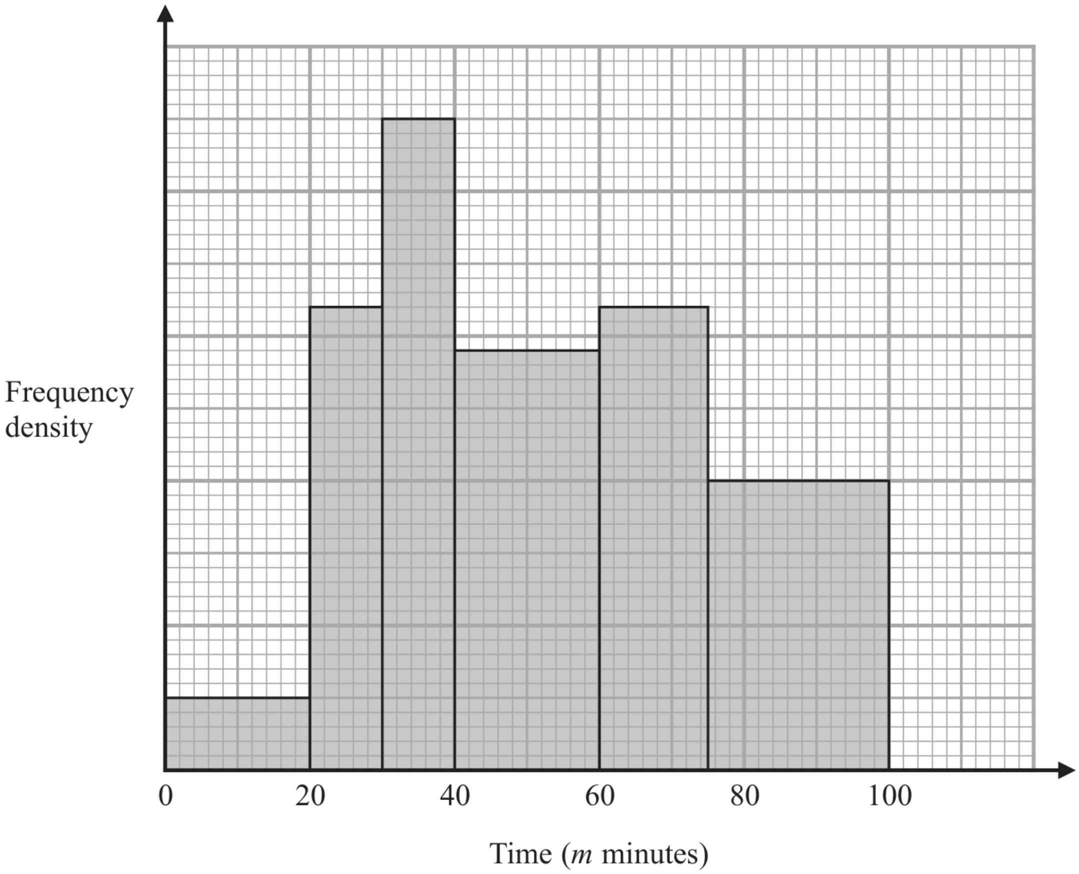
Question 14

4MA1-1H Q21 Q21

MODULAR UNIT 1

3 marks

21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$
There are 10 children for whom $m \leqslant 20$

Work out an estimate for the number of children for whom $50 < m \leqslant 80$

....

Question 15

4MA1-1H Q24 Q24

MODULAR UNIT 1
4 marks

24 Express each of a, b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$a =$ ....

$b =$ ....

$c =$ ....

WORKING SPACE

Question 16
4MA1-2H Q3 Q3

MODULAR UNIT 1
3 marks

3 An aeroplane travelled from New York City to Los Angeles.

The aeroplane travelled a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the aeroplane.
Give your answer in kilometres per hour correct to the nearest whole number.

.... kilometres per hour

WORKING SPACE

Question 17

4MA1-2H Q4 Q4

MODULAR UNIT 1
3 marks

4 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

WORKING SPACE

Question 18

4MA1-2H Q5 Q5

MODULAR UNIT 1
4 marks

5 The diagram shows an 8-sided shape *ABCDEFGH*.

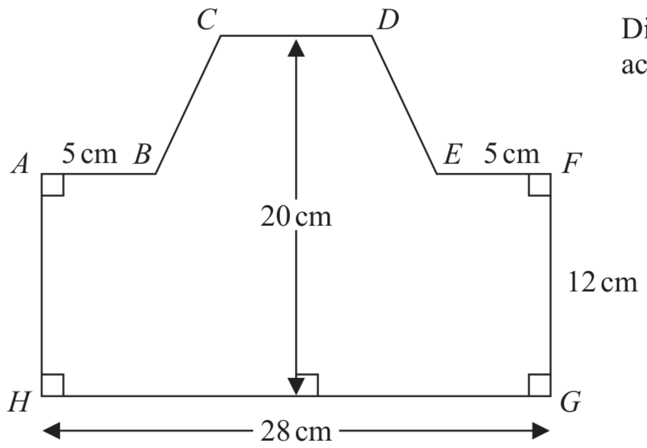


Diagram **NOT** accurately drawn

$HG = 28$ cm $FG = 12$ cm $AB = EF = 5$ cm
The height of the shape is 20 cm
CD is parallel to HG

The area of shape *ABCDEFGH* is 434 cm^2

Find the length of *CD*.

.... cm

Working space for Question 18

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 19

4MA1-2H Q6 Q6

MODULAR UNIT 1

3 marks

6 The diagram shows triangle *PQR*.

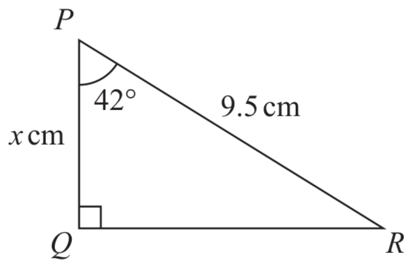


Diagram **NOT** accurately drawn

Work out the value of x

Give your answer correct to one decimal place.

$x =$ ....

WORKING SPACE

Question 20

4MA1-2H Q7 Q7

MODULAR UNIT 1
3 marks

7 Change a speed of 81 kilometres per hour to a speed in metres per second.

.... metres per second

WORKING SPACE

Question 21

4MA1-2H Q11 Q11(i), Q11(ii)

MODULAR UNIT 1

3 marks

11 (i) Factorise $x^2 + 5x - 24$

....

(2)

(ii) Hence, solve $x^2 + 5x - 24 = 0$

....

(1)

WORKING SPACE

Question 22

4MA1-2H Q14 Q14(a)

MODULAR UNIT 1
3 marks

14 (a) Expand and simplify $(5 - x)(2x + 3)(x + 4)$
Show your working clearly.

....

(3)

WORKING SPACE

Question 23

4MA1-2H Q15 Q15(a)

MODULAR UNIT 1
4 marks

15 (a) Solve $\dfrac{4x+5}{3} - \dfrac{3-2x}{2} = 13$

Show clear algebraic working.

$x =$ ....

(4)

WORKING SPACE

Question 24

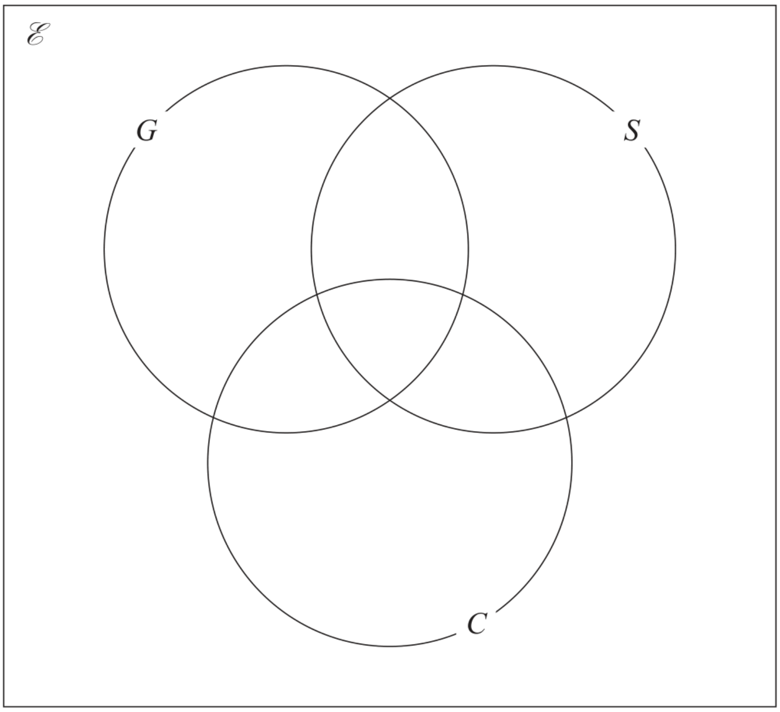
4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

MODULAR UNIT 1
8 marks

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers
31 have sheep
53 have chickens
6 have goats, sheep and chickens
11 have sheep and goats
17 have sheep and chickens
18 have goats and chickens
20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

(b) Find

Question 24 (continued)

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

MODULAR UNIT 1
8 marks

(i) $n(G)$

....
(1)

(ii) $n([G \cup S]')$

....
(1)

(iii) $n(G' \cap C)$

....
(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

....
(2)

Question 25

4MA1-2H Q18 Q18

MODULAR UNIT 1

3 marks

18 $X = \dfrac{2a - b}{f}$

$a = 7.5$ correct to 1 decimal place.
$b = 3.42$ correct to 2 decimal places.
$f = 2$ correct to the nearest whole number.

Work out the upper bound of the value of X
Show your working clearly.

....

WORKING SPACE

Question 26

4MA1-2H Q19 Q19

MODULAR UNIT 1
3 marks

19 $a = \dfrac{14}{3x - 7}$ $\qquad x = \dfrac{7}{4y - 3}$

Express a in the form $\dfrac{py + q}{ry + s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a =$ ....

WORKING SPACE

Question 27
4MA1-2H Q20 Q20

MODULAR UNIT 1
4 marks

20 The diagram shows four identical circles drawn inside a square.

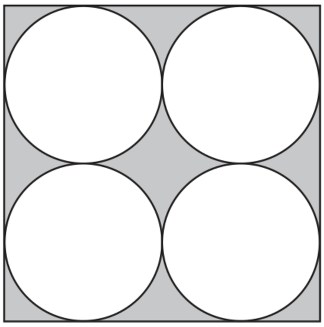


Diagram **NOT** accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.
Give your answer correct to 3 significant figures.

.... cm

Working space for Question 27

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 28

4MA1-2H Q22 Q22

MODULAR UNIT 1
7 marks

22 $ABCD$ is a kite, with diagonals AC and BD, drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D.

(.............................. , ...)

(.............................. , ...)

WORKING SPACE

Question 29
4MA1-2H Q24 Q24

MODULAR UNIT 1
3 marks

24

$$\frac{18 \times \left(\sqrt{27}\right)^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n
Show your working clearly and simplify your expression.

$x =$ ....

WORKING SPACE